Zero Features, Full Signal: Topology-Driven Coordination Analysis in Meme-Token Markets

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***Abstract*—Coordinated wallet groups drive the majority of manipulation in Solana's meme-token markets, yet existing detection methods rely on hand-crafted token-level features that cannot capture the relational structure of coordination itself. We model the ecosystem as a heterogeneous wallet–token interaction graph and investigate how much of the information needed to identify and anticipate coordinated manipulation is encoded in interaction topology alone. *First*, a featureless heterogeneous GAT using only learnable type embeddings *detects* coordination with AUC = 0.867 [0.852, 0.882], outperforming five tabular baselines operating on 116 engineered features (best: Random Forest, 0.855). *Second*, a Bochner-encoded temporal extension *predicts* where coordinated wallets will target next, achieving MRR = 0.490 and Hits@10 = 0.952 ($2.7\times$ random), outperforming EdgeBank and popularity heuristics that score below chance. *Third*, unsupervised clustering of the learned embeddings reveals three coordination *archetypes*, bundled bot operators, organic retail, and loosely synchronised groups, without any explicit supervision on wallet types. An early-detection analysis shows that topological signal requires approximately three days of accumulated interactions to match feature-based methods, establishing the two modalities as complementary for staged monitoring.**

I. Introduction

Coordination is the organising principle behind manipulation in meme-token markets. On Solana, platforms such as Pump.fun launch upwards of 50,000 tokens per day, collectively generating over $5 billion in daily DEX volume, and a majority are associated with orchestrated activity: insider wallets inflate early volume to attract retail capital, then withdraw liquidity [1]. The MemeTrans dataset documents over 41,000 such tokens, of which approximately 79 % are labelled high-risk based on return ratios. What makes these schemes effective is not any single wallet's behaviour but the *coordination* among wallets: same-transaction bundling, sub-second co-trading across multiple tokens, and hub-and-spoke subgraph formation. These are inherently relational signals that per-token aggregate statistics can only approximate.

Graph topology captures coordination in a way that token-level features cannot. Coordinated clusters participate across *multiple* tokens, and a graph representation preserves the full neighbourhood structure, capturing who traded what, when, and alongside whom, enabling detection of coordination patterns that persist across tokens and across time.

Prior work exists along three separate strands that have not yet been integrated. On Solana, MemeTrans [1] engineers 122 features from on-chain data and applies gradient-boosted trees; SolRugDetector [2] compiles over 76,000 rug-pull tokens from on-chain transaction patterns; and Alizadeh and Khabbazian [3] analyse Solana NFT graph structure without GNNs. None of these Solana-specific works employ graph neural networks or temporal message passing.

On Ethereum, TokenScout [4] and RugScreener [5] apply temporal GNNs to token fraud, while Mazorra et al. [6] provide an early ML baseline. Weber et al. [7] applied GCNs to Bitcoin, and Li et al. [8] survey blockchain GNN anomaly detection. All rely on attributed graphs with substantial node features.

A parallel line of work develops temporal GNN primitives: TGAT [9] introduced Bochner time encodings, TGN [10] combined memory with temporal attention, and Poursafaei et al. [11] established rigorous evaluation protocols for dynamic link prediction.

Despite this body of work, three strands remain disconnected: no prior study applies temporal GNNs to Solana meme-token coordination, no study asks whether topology alone, without hand-crafted features, suffices for risk detection, and no study formulates wallet-token interaction prediction as temporal link prediction on blockchain graphs.

We address this gap with a single research question: *how much of the information needed to identify and anticipate coordinated manipulation is encoded in wallet–token interaction topology, independent of hand-crafted features?* We answer under three aspects:

(i) *Detection.* Graph topology *detects* coordination: a featureless HeteroGAT achieves AUC = 0.867, exceeding the best tabular baseline (0.855) that operates on 116 engineered features.

(ii) *Prediction.* Graph topology *predicts* where coordinators will target next: a temporal extension achieves MRR = 0.490 for next-token prediction.

(iii) *Characterisation.* Topology reveals coordination *archetypes*: unsupervised HDBSCAN clustering [12] discovers bundled bot operators, organic retail, and loosely synchronised groups from embeddings alone.

II. Methodology

A. Problem Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a heterogeneous graph with node types $\mathcal{T} = \{\texttt{token}, \texttt{wallet}\}$ and edge types $\mathcal{R} = \{\texttt{trade}, \texttt{reverse_trade}, \texttt{co_trade}, \texttt{same_tx}\}$. Each token node τ has a binary label $y_\tau \in \{0, 1\}$ indicating high-risk status. The co-trading and same-transaction edges encode two complementary facets of coordination: implicit synchronisation (wallets independently arriving at the same tokens within seconds) and explicit bundling (wallets transacting within the same atomic on-chain operation).

We study coordination through two complementary tasks. *Task 1 (Retrospective detection):* Given $\mathcal{G}$ and labels on the

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training split, predict $\hat{y}_\tau$ for tokens in the test split, identifying tokens whose trading activity bears the structural fingerprint of coordination. *Task 2 (Prospective prediction):* Given a sequence of timestamped interactions $\{(w_i, \tau_j, t_{ij})\}_{t_{ij} \le t_q}$, predict which token τ wallet w will interact with at time $t > t_q$, forecasting where coordinated groups will converge next.

B. Data and Graph Construction

We draw from the MemeTrans dataset [1], which catalogues meme-token activity on Solana's Pump.fun launchpad. The dataset provides per-token features (holder concentration, price dynamics, trading volume statistics) and a three-class risk label (`high`, `medium`, `low`) derived from return ratios; we binarise this as y_τ=1 for `high` and y_τ=0 otherwise, following the original authors' primary partition. We extract a three-week subset (December 9–29, 2024) comprising 65.5 million transactions queried from Google BigQuery's Solana public dataset and split chronologically: week 50 (Dec 9–15) for training, week 51 (Dec 16–22) for validation, week 52 (Dec 23–29) for testing. Each split builds an independent graph to prevent temporal leakage [11]. Table I reports split statistics.

Exploratory analysis. The label distribution is heavily skewed: approximately 79 % of labelled tokens are classified as high-risk across all three splits (Table I), with a stable ratio of ∼3.7:1 that confirms the prevalence of manipulation on Pump.fun. Weekly transaction volume is stable (∼20–23M per week), as are active wallet counts (42–46K) and edge counts, indicating that the three-week window captures a stationary regime of market activity rather than a transient event.

The bipartite degree distribution reveals a counter-intuitive asymmetry: *low-risk* tokens attract substantially more wallets (median in-degree ∼1,174 in training) than high-risk tokens (median ∼380). Legitimate tokens accumulate organic trading volume over their lifetime, whereas coordinated pump-and-dump tokens exhibit intense but concentrated activity from a smaller set of wallets before liquidity is withdrawn. This inversion means that raw wallet count is not a reliable risk indicator—coordination *structure* (who trades alongside whom) matters more than coordination *volume*, motivating the graph-based approach. Wallet out-degree is right-skewed: the median active wallet trades 29 distinct tokens (the minimum is 20 by construction), while the 95th percentile reaches ∼105 tokens and the most active wallet trades over 2,600. This heavy tail captures the most prolific coordinators, whose trading breadth generates the densest co-trading subgraphs.

Data cleaning and preprocessing. We retain only *active* wallets, defined as those trading ≥20 distinct tokens. This threshold represents a coverage–precision trade-off: a stricter filter (≥50 tokens) yields AUC = 0.896 on a denser subgraph but retains only ∼1.7 % of wallets, discarding those involved in narrowly-targeted campaigns; lowering the threshold below 20 admits wallets whose co-trading edges reflect chance overlap rather than systematic coordination. We select ≥20 as the operating point that maximises population coverage (∼4 % of wallets, responsible for the majority of inter-token coordination) while preserving structural distinctiveness. MemeTrans provides

TABLE I: Dataset statistics per chronological split. High-risk tokens outnumber low-risk by ∼3.7:1 across all splits.

	Train	Val	Test
Transactions	22.6M	22.5M	20.4M
Active wallets	42,233	46,082	44,661
Labelled tokens	2,679	2,906	2,538
High-risk	2,130	2,278	1,995
Low-risk	549	628	543
W→T edges	1.81M	1.97M	1.97M
Co-trade edges	634K	691K	670K
Same-tx edges	59K	53K	51K

122 raw features per token (holder Gini coefficients, top-k holder percentages, price and volume statistics, temporal indicators); after removing identifier columns and label-leaking fields, 116 features remain for tabular baselines. The featured HeteroGAT uses these 116 token features plus 15 engineered wallet-level behavioural features (transaction counts, temporal statistics, token breadth), yielding 131 input features total. All features are z-score normalised (fit on training data, applied to all splits, clipped to $[-10, 10]$ to limit outlier influence).

The graph encodes coordination through two dedicated edge types. *Co-trading edges* connect wallet pairs with a temporally-decayed weight measuring synchronised activity:

$$e_{ab} = \sum_{m \in \mathcal{M}_a \cap \mathcal{M}_b} \exp\!\big(-\lambda\, |t_a^{(m)} - t_b^{(m)}|\big), \tag{1}$$

where $t_a^{(m)}$ is wallet a's first trade of token m and λ=$1/3600\,\mathrm{s}^{-1}$ (half-life $t_{1/2}$≈42 min), chosen to distinguish sub-hour coordination from organic co-trading patterns. A binary-search threshold θ on the aggregated weight (θ≈60) is calibrated on the training graph to achieve a target mean co-trading degree of 15, yielding ∼660K directed edges. *Same-transaction edges* (binary) connect wallets co-occurring in a single on-chain transaction signature, capturing bundled bot activity (∼55K edges). These two edge types encode a coordination spectrum: same-tx edges capture *hard* coordination (wallets bundled by a single program), while co-trading edges capture *soft* coordination (wallets arriving independently but within seconds). Trading edges connect each wallet to every token it traded (∼1.9M per split), forming the bipartite backbone through which even indirect coordination propagates via two-hop paths (wallet→token→wallet).

C. Coordination-Aware Architecture

The featureless HeteroGAT has ∼69K parameters. Each node receives a learnable type embedding $\mathbf{h}_i^{(0)} = \mathbf{e}_{\tau(i)} \in \mathbb{R}^d$ (d=64). No other input is provided; the model must infer coordination entirely from graph topology.

Proposition (GAT degeneracy). *If* $\mathbf{h}_j^{(0)}$=$\mathbf{c}$ *for all* j *of the same type, then the GAT attention coefficients* (3) *reduce to* $\alpha_{ij}^{(r)} = 1/|\mathcal{N}_r(i)|$ *for every neighbour* j.

Proof sketch. With identical inputs, $\mathbf{W}_r\mathbf{h}_j^{(1)}$=$\mathbf{W}_r\mathbf{c}$ for all j, so $\mathbf{a}_r^\top[\mathbf{W}_r\mathbf{c}\|\mathbf{W}_r\mathbf{c}]$ is a constant independent of j. The softmax over identical logits yields $\exp(e)/\sum\exp(e) = 1/|\mathcal{N}_r(i)|$, and attention becomes uniform averaging, equivalent to mean pooling [13]. All degree information is discarded.

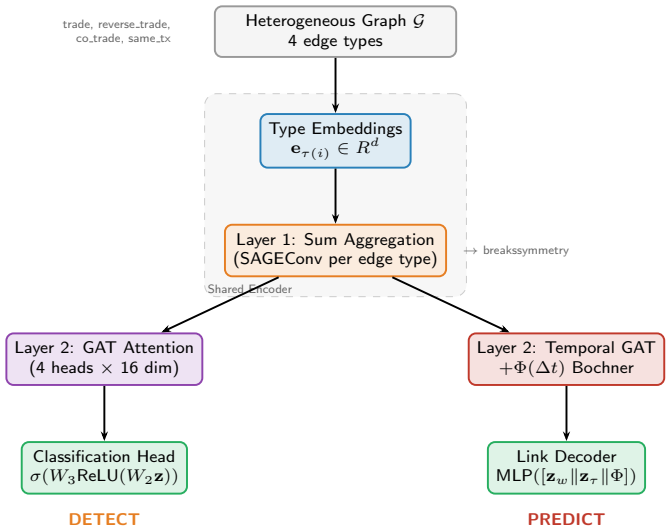

Fig. 1: Overview of the coordination detection pipeline. Wallet–token interactions are encoded as a heterogeneous graph with four edge types. The featureless HeteroGAT (left) detects coordination retrospectively; the temporal extension (right) predicts where coordinators will target next.

We resolve this with a two-stage *sum-then-attention* design. Layer 1 uses sum aggregation [14] to encode coordination degree:

$$\mathbf{h}_i^{(1)} = \mathrm{ELU}\Big(\mathrm{LN}\Big(\sum_r \sum_{j\in\mathcal{N}_r(i)} \mathbf{W}_r^{(1)}\,\mathbf{h}_j^{(0)}\Big)\Big), \tag{2}$$

where $\mathbf{W}_r^{(1)} \in \mathbb{R}^{d\times d}$ (d=64), r indexes edge type, and LN denotes LayerNorm. A token traded by 100 wallets now has embedding magnitude 100× that of one traded by a single wallet, encoding the scale of coordination activity.

Layer 2 applies multi-head GAT attention [15] (4 heads, 16 dims each) on the now-differentiated nodes:

$$\alpha_{ij}^{(r)} = \frac{\exp\big(\mathrm{LReLU}(\mathbf{a}_r^\top[\mathbf{W}_r\mathbf{h}_i^{(1)}\|\mathbf{W}_r\mathbf{h}_j^{(1)}])\big)}{\sum_{k\in\mathcal{N}_r(i)}\exp\big(\mathrm{LReLU}(\mathbf{a}_r^\top[\mathbf{W}_r\mathbf{h}_i^{(1)}\|\mathbf{W}_r\mathbf{h}_k^{(1)}])\big)}. \tag{3}$$

Messages from different edge types arriving at the same node are aggregated by summation:

$$\mathbf{z}_i = \mathrm{ELU}\Big(\sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_r(i)} \alpha_{ij}^{(r)}\,\mathbf{W}_r^{(2)}\mathbf{h}_j^{(1)}\Big) + \mathbf{h}_i^{(1)}, \tag{4}$$

where the residual connection preserves the degree-differentiated representations from Layer 1. The summation over edge types is a deliberate design choice: it allows the attention mechanism within each relation to weight individual neighbours, while preserving the additive contribution of each coordination channel: a wallet receiving both co-trading and same-tx messages accumulates evidence from both sources. A two-layer MLP (64 → 32 → 1) produces token-level predictions $\hat{y}_i = \sigma(W_3\,\mathrm{ReLU}(W_2\,\mathbf{z}_i^{\mathrm{tok}}))$. Training minimises the weighted binary cross-entropy:

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{N}\sum_{i=1}^{N}\big[w_+\,y_i\log\hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\big], \tag{5}$$

where $w_+ = n_{\mathrm{neg}}/n_{\mathrm{pos}} \approx 0.26$ down-weights the majority high-risk class. We use AdamW (lr=10^{-3}, wd=10^{-3}), cosine annealing ($T_{\max}$=300), gradient clipping (max norm 1.0), and early stopping (patience 30 on validation AUC). The model reaches peak validation AUC = 0.856 at epoch 28; training loss continues to decrease thereafter while validation plateaus, the classic overfitting signature. Early stopping terminates at epoch 58; inference over the full test graph completes in <1 s, confirming the model's suitability for deployment.

TABLE II: Parameter count by component for both model variants.

Component	Featureless	Featured
Type/Feature embeddings	128	16,640
SAGEConv L1 (4 edge types)	33,024	33,024
GATConv L2 (4 × 4 heads)	33,792	33,792
Prediction head (64 → 32 → 1)	2,113	2,113
Total	**69,057**	**85,569**

We also train a featured variant that replaces type embeddings with MLP projections of the raw features (token: 116 → 64 → 64; wallet: 15 → 64 → 64), keeping the same architecture (∼86K parameters), to isolate the contribution of features on top of topology. Table II details the parameter budget.

D. Temporal Extension

Having shown that topology detects coordination retrospectively, we now ask whether it can predict where coordinators will target next. The temporal model retains the same two-layer structure (Layer 1 SAGEConv, $\mathbb{R}^{128} \to \mathbb{R}^{128}$, sum aggregation, followed by Layer 2 attention) but replaces the static GAT in Layer 2 with a *temporal* attention mechanism that conditions on the time gap between interactions (∼314K parameters total). Specifically, we augment the attention with a Bochner time encoding [9]. For each edge with time gap $\Delta t = (t_{\mathrm{query}} - t_{\mathrm{edge}})/86400$ (normalised to days):

$$\Phi(\Delta t) = \frac{1}{\sqrt{d_\Phi}}\big[\cos(\omega_1\Delta t + b_1), \ldots, \cos(\omega_{d_\Phi}\Delta t + b_{d_\Phi})\big], \tag{6}$$

where ω_k, b_k are learnable (d_Φ=64). The construction follows from Bochner's theorem [16]: any shift-invariant positive-definite kernel $k(\Delta t)$ admits a spectral representation $k(\Delta t) = \mathbb{E}_\omega[\cos(\omega\Delta t)]$, so the inner product $\langle\Phi(t_1), \Phi(t_2)\rangle$ approximates a valid temporal kernel. Making frequencies learnable lets the model discover task-relevant timescales automatically [9].

The temporal attention logits incorporate this encoding:

$$e_{ij}^{(r)} = \frac{\mathrm{LReLU}\big(\mathbf{a}_r^\top[\mathbf{W}_s\mathbf{h}_i\|\mathbf{W}_d\mathbf{h}_j\|\Phi(\Delta t_{ij})]\big)}{\sqrt{2d + d_\Phi}}. \tag{7}$$

For a candidate interaction (w, τ) at time t, the decoder scores:

$$\hat{y}_{w\tau}(t) = \sigma\big(\mathrm{MLP}([\mathbf{z}_w\|\mathbf{z}_\tau\|\Phi(t - t_w^{\mathrm{last}})\|\Phi(t - t_\tau^{\mathrm{last}})])\big), \tag{8}$$

where $\mathbf{z}_w, \mathbf{z}_\tau$ are node embeddings after two layers of temporal HeteroGAT. For each observed edge (w, τ^+, t) we sample K=19 negative destinations $\{\tau_k^-\}$ uniformly at random and minimise the binary cross-entropy:

$$\mathcal{L}_{\mathrm{link}} = -\log\hat{y}_{w\tau^+} - \frac{1}{K}\sum_{k=1}^{K}\log(1 - \hat{y}_{w\tau_k^-}), \tag{9}$$

following the fixed negative sampling protocol of [11]. The model has ∼314K parameters (265K encoder + 49K decoder).

E. Baselines and Evaluation

For classification, we compare against five feature-based models operating on the 116 cleaned token features. The *null baseline* is L2-regularised logistic regression with balanced class weights, the simplest linear model that preserves all input features and provides a lower bound on learnable signal. We also evaluate SVM with RBF kernel, Random Forest (200 trees), Gradient Boosting (200 estimators), and MLP ($116 \to 256 \to 128 \to 1$, $\sim$64K parameters). For link prediction, we compare against three baselines: (i) a random baseline (expected MRR ≈ 0.180 for 20 candidates), (ii) EdgeBank [11] (memorises training edges and predicts repeats), and (iii) a most-popular-token heuristic that ranks candidates by normalised training frequency. All reported metrics are evaluated out-of-sample on the held-out week-52 test set unless otherwise noted.

Classification metrics: AUC-ROC (primary), F1, AP, precision, and recall, with 95 % bootstrap confidence intervals (n=1,000) and McNemar's test for pairwise significance. Link prediction uses Mean Reciprocal Rank and Hits@K:

$$\text{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\text{rank}_i}, \quad \text{Hits@}K = \frac{1}{N}\sum_{i=1}^{N}\mathbb{1}[\text{rank}_i \le K], \tag{10}$$

where rank_i is the position of the true destination among $1+K_{\text{neg}}$ candidates (1 positive + 19 negatives).

Algorithm 1 summarises the full pipeline.

III. Results

A. Coordination Detection

Table III reports test-set classification performance. The featureless HeteroGAT (AUC = 0.867) outperforms all five tabular baselines despite using zero hand-crafted features, demonstrating that graph topology alone encodes coordination signals that 116 engineered features approximate imperfectly. When topology and features are combined, the featured HeteroGAT reaches the highest overall score (AUC = 0.916). Decomposing this gain is instructive: adding graph structure to features yields ΔAUC$= +0.061$ (RF 0.855 $\to$ featured 0.916), whereas adding features to topology yields only ΔAUC$= +0.049$ (featureless 0.867 $\to$ featured 0.916). Graph structure therefore contributes more marginal discriminative power than 116 engineered features.

These performance differences are not attributable to sampling variance: McNemar's test confirms significance for the featureless HeteroGAT against all individually tested baselines—logistic regression (χ^2=13.6, p<0.001), MLP (χ^2=7.2, p=0.007), and Random Forest (χ^2=7.3, p=0.007). The featureless model's optimal threshold is 0.79, compared to 0.09 (LogReg) and 0.13 (MLP), indicating well-calibrated probabilities.

Error analysis. False negatives concentrate on tokens with fewer than five connected wallets, where the coordination subgraph is too sparse for two-hop message passing. False positives arise on organically popular tokens whose high connectivity mimics coordinated structure. The 79 % high-risk base rate makes threshold-invariant AUC-ROC the appropriate primary metric.

Algorithm 1: Coordination Detection Pipeline

Input: Weekly parquet files $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}$; feature matrix $\mathbf{X}$; labels $\mathbf{y}$
Output: Trained models, predictions, evaluation metrics

```
/* Phase 1: Graph construction                          */
foreach split s ∈ {train, val, test} do
    Load transactions T_s from parquets
    Build wallet→token edges from T_s
    Compute temporally-decayed co-trade edges (Eq. 1)
    Extract same-transaction edges
    Assemble G_s as PyG HeteroData
Z-score normalise features (fit on train)
/* Phase 2: Coordination detection (classification)     */
Initialise FeaturelessHeteroGAT (Eqs. 2–3)
w_+ ← n_neg/n_pos ≈ 0.26                  // class weight
for epoch = 1, ..., 300 do
    ŷ ← forward pass on G_train
    L ← BCE(ŷ, y; pos_weight=w_+)
    Update via AdamW, cosine annealing, grad clip 1.0
    if val AUC not improved for 30 epochs then
        break
/* Phase 3: Coordination prediction (link prediction)   */
Extract chronological events (w, τ, t) from T_train
Encode G_train with temporal HeteroGAT (Eqs. 6–7)
for each batch of events do
    Sample 19 negative tokens per positive
    Score via link decoder (Eq. 8)
    L ← BCE(pos scores, neg scores)
    Update decoder parameters
/* Phase 4: Evaluation                                  */
Compute AUC, F1, AP with 95% bootstrap CIs
Compute MRR, Hits@K for link prediction
Run early detection at h ∈ {1, 6, 24, 72, 168} hours
```

B. Edge-Type Ablation

The featureless model's reliance on topology raises a natural follow-up: which edge types carry the coordination signal? To answer this, we ablate the graph by removing one edge type at a time and retraining from scratch (Table IV).

Same-transaction edges are the dominant coordination signal: removing them drops AUC by 0.047 (from 0.867 to 0.820), a 5.4% relative decrease. This is consistent with same-tx edges encoding the strongest form of coordination, wallets literally transacting within a single atomic operation. By contrast, removing co-trading edges has negligible impact (ΔAUC$= +0.001$), suggesting that the temporal co-trading signal is largely redundant with information already captured by the bipartite trading structure. Even removing *all* wallet–wallet edges produces only a 0.004 AUC drop, indicating that the bipartite wallet→token→wallet message-passing path recovers most coordination information indirectly.

C. Coordination Prediction

Having established that topology detects coordination, we now ask whether it can predict where coordinated wallets will converge next. Table V reports link prediction performance, where both heuristic baselines perform *worse than random*.

TABLE III: Test-set classification performance (week 52). All models use the same chronological train/val/test split. All models report 95 % bootstrap CI (n=1,000). Best result per column in bold. The graph-based models (bottom) use heterogeneous message passing over four edge types; the featureless variant receives zero hand-crafted features.

	Model	Features	Parameters	AUC-ROC	F1	AP	Precision	Recall
Tabular	Logistic Regression	116	—	.841 [.825, .857]	.887	.955	.812	**.977**
	SVM (RBF kernel)	116	—	.834 [.817, .849]	.805	.953	**.942**	.703
	Random Forest	116	—	.855 [.840, .870]	.878	.960	.864	.893
	Gradient Boosting	116	—	.849 [.835, .863]	.878	.959	.865	.890
	MLP	116	64K	.850 [.835, .865]	.890	.958	.827	.963
Graph	FeaturelessHeteroGAT*	0	69K	.867 [.852, .882]	.899	.960	.870	.931
	FeaturedHeteroGAT	131‡	86K	**.916** [.905, .926]	**.918**	**.977**	.909	.927

*Across seeds $\{42, 123, 7\}$: AUC = 0.869 ± 0.002, confirming training stability.
‡116 token + 15 wallet features.

TABLE IV: Edge-type ablation: featureless HeteroGAT retrained on restricted graphs. Same-tx edges are the critical coordination signal.

Configuration	AUC	Δ	AP
Full graph (4 types)	.867	—	.960
− co-trade edges	.868	+.001	.961
− all coordination (W-W)	.863	−.004	.958
− same-tx edges	.820	−.047	.929

TABLE V: Temporal link prediction (within-week validation, 1 positive + 19 negatives per query). EdgeBank [11] memorises training edges; Popularity ranks by token frequency. Both heuristics score below random, confirming that the temporal model learns coordination patterns rather than recurrence or popularity.

Model	MRR	Hits@1	Hits@3	Hits@10
Random	.180	.050	.150	.500
EdgeBank	.056	.005	.007	.009
Popularity	.056	.002	.005	.024
Cosine Sim.	.250	.078	.268	.695
Temporal HeteroGAT	**.490**	**.296**	**.581**	**.952**

EdgeBank [11] (MRR = 0.056) fails because meme-token trading is novelty-driven: wallets constantly interact with new tokens rather than revisiting past ones, so memorising historical edges is actively misleading. The popularity baseline (MRR = 0.056) fails because insider coordination targets niche tokens, not popular ones. A cosine-similarity baseline using the same frozen embeddings achieves MRR = 0.250, confirming that the temporal decoder contributes substantially beyond static embedding proximity. The temporal HeteroGAT achieves MRR = 0.490 (2.7× random, ∼8.8× EdgeBank), placing the true next token at rank ∼2 on average among 20 candidates. Hits@10 = 0.952 means the true token is in the top half 95 % of the time. These results confirm that the temporal model learns structural coordination patterns, specifically which wallet clusters converge on which tokens, rather than exploiting recurrence or popularity.

These results have a direct operational consequence: predicted convergence of wallets historically associated with manipulation constitutes an early warning that retrospective classification cannot provide, bridging the temporal gap identified in Section III-D. The learnable Bochner frequencies adapt to Solana coordination timescales (sub-minute bundling, hourly organic cycles, daily rhythms) without manual specification. The reported MRR of 0.490 uses frozen GNN embeddings and is therefore a lower bound on temporal predictive capacity.

TABLE VI: Early detection: classification AUC as a function of observation time. Tabular baselines use full-week features (upper bound).

Model	1h	6h	24h	3d	7d
LogReg	.841	.841	.841	.841	.841
RF	.855	.855	.855	.855	.855
Feat.less GAT	.500	.500	.542	.611	.841
% edges	0.1	1	8	34	95

D. Signal Emergence

A deployment-critical question is how quickly topological signal becomes useful. We evaluate the featureless HeteroGAT on subgraphs containing only edges from the first h hours of the test week; tabular baselines use features computed from the full week and are therefore time-invariant upper bounds (Table VI).

At 1 hour (0.1 % of edges), coordination structure is too sparse for the GNN (AUC = 0.500). Signal emerges at 24 hours (0.542, 8 % of edges) and strengthens at 3 days (0.611, 34 %), the critical mass at which two-hop coordination paths become distinguishable from chance. At 7 days the GNN converges to AUC = 0.841. Feature-based methods provide immediate signal; graph topology requires edge accumulation but converges without feature engineering. A practical system would combine both: tabular features for instant screening, augmented by evolving graph structure—where the link predictor (Section III-C) bridges the accumulation gap.

E. Coordination Archetypes

To examine whether the learned embeddings encode interpretable structure, we apply HDBSCAN clustering [12] to the 64-dimensional wallet embeddings (minimum cluster size 50, Euclidean distance), obtaining 25 clusters with only 2.1 % noise despite no supervision on wallet types.

Three coordination archetypes emerge. *(a) Tight, high-coordination wallets* (∼12 % of the active population) exhibit narrow token focus and dense same-transaction connectivity, consistent with automated bundler operations [1]. These wallets are disproportionately linked to high-risk tokens (34 % of high-risk connections vs. 12 % population share). *(b) Broad, low-coordination wallets* (∼45 %) trade across many tokens with minimal same-transaction overlap, interacting with high-risk tokens at the population base rate (∼79 %), consistent with organic retail behaviour. *(c) Intermediate, loosely synchronised*

wallets display elevated temporal co-trading similarity but few same-transaction edges—the "soft coordination" that the temporally-decayed co-trading edges (Eq. 1) are designed to capture and that same-transaction analysis alone would miss.

The separation between archetypes reinforces the classification and prediction results: the coordination edges that drive detection and prediction also produce semantically meaningful clusters, confirming that the learned representations encode genuine coordination structure rather than statistical artefacts.

IV. Discussion

The three experimental views—detection, prediction, and characterisation—converge on a single conclusion: graph topology functions as a coordination fingerprint. The featureless model's superiority over all tabular baselines demonstrates that wallet–token interaction topology directly encodes coordination signals that 116 engineered aggregate statistics can only approximate.

This claim warrants nuance. The AP scores are effectively identical between the featureless model (0.960) and Random Forest (0.960), indicating comparable positive-class ranking quality. The AUC-ROC advantage ($+0.012$) reflects stronger discrimination in the low false-positive-rate regime—where a monitoring system would operate, since early warnings require high specificity. The more important finding is the marginal contribution analysis: adding graph structure to features yields $\Delta\text{AUC} = +0.061$, whereas adding features to topology yields only $\Delta\text{AUC} = +0.049$. Topology and features capture partially overlapping but distinct signals, and their combination ($\text{AUC} = 0.916$) substantially exceeds either alone.

The temporal extension further establishes that this fingerprint is *predictive*, not merely retrospective. The failure of recurrence- and popularity-based heuristics—scoring below chance—confirms genuine structural learning. The cosine-similarity baseline ($\text{MRR} = 0.250$ vs. temporal model's 0.490) isolates the contribution of Bochner-encoded temporal attention: static embedding proximity accounts for roughly half the signal, while time-aware attention contributes the remainder by learning coordination timescales. The sum-then-attention design is architecturally necessary: standard GAT with identical embeddings reduces to mean pooling, discarding degree information. This principle generalises to any featureless GNN setting. The early-detection analysis establishes the two modalities as complementary, defining a staged architecture in which the link predictor bridges the initial accumulation period.

Several limitations qualify these findings: (i) weekly snapshot granularity that sacrifices intra-day dynamics, which continuous-time models [9], [10], [17] could capture; (ii) a high positive rate (79 %) that inflates absolute metrics while preserving relative comparisons; (iii) pre-computed GNN embeddings for link prediction that prevent end-to-end gradient flow ($\text{MRR} = 0.490$ is a lower bound); and (iv) only 28.8 % of test-week wallets appear in training—the remaining 71.2 % receive identical type embeddings, discarding historical reputation. *Persistent wallet embeddings*, carrying forward learned representations across weeks, would let the GNN recognise returning coordinators immediately—integrating naturally with TGN-style per-node memory [10]. End-to-end temporal training, cross-week stability analysis, and entity resolution of co-clustered wallets remain as future directions.

Taken together, these findings establish that graph topology provides a self-contained coordination signal: the featureless approach requires only the public transaction ledger, making detection, prediction, and characterisation accessible without proprietary data or feature engineering infrastructure.

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